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DAMPING RING ASSEMBLY

Abstract

A damping ring assembly may include a plurality of damping members, each of the damping members including an arc length, a radial thickness, a plurality of recesses located on an end surface and a retention groove on an outer surface of the damping members. A damping ring assembly may also include a plurality of brackets for attaching the plurality of damping members to a circumferential surface of a drive sprocket, the plurality of brackets each including a radially inner leg, a radially outer leg, and a joining central portion that connects the radially inner leg and the radially outer leg.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to a track drive assembly and continuous ground-engaging track for a mobile industrial vehicle, and more particularly, to a sprocket damping assembly for a drive sprocket and the ground-engaging track.

BACKGROUND

[0002] Mobile industrial machines, such as track-type tractors, are used in a variety of applications and offer the advantage of a rugged build and performance in challenging ground conditions on a work site. An undercarriage of a track-type industrial machine may include an endless track made of a number of connected links, a drive sprocket driven by a power source of the industrial machine (e.g., an internal combustion engine or electric motor) that meshes with the links to drive the track, and a number of idler wheels that distribute the weight of the work machine along the track.

[0003] Track-type industrial machines may generate loud noises during operation, which may be objectionable when operating in highly populated areas. Environmental regulations, both in the United States and in other countries, are increasingly being directed to suppressing noise generated by the operation of construction equipment. One potential source of operating noise that can also be a source of wear on the industrial machine is the metal-to-metal impact occurring between the teeth of the drive sprocket and bushings of the track links during the meshing between the teeth and the links.

[0004] A sprocket damping assembly for a sprocket wheel is described in U.S. Pat. No. 11,535,319 B2, issued Dec. 27, 2022 (“the ‘319 Patent”). The sprocket damping system described in the ‘319 Patent includes a damping ring that is disposed on the sprocket wheel, and the damping ring includes an integrally formed retention groove for retaining a retention device. While the sprocket damping assembly described in the ‘319 Patent may be helpful in some circumstances, the sprocket damping assembly may be subject to movement after being secured to the sprocket wheel.

[0005] Aspects of the present disclosure may solve one or more of the problems set forth above and/or other problems in the art. The scope of the current disclosure, however, is defined by the attached claims, and not by the ability to solve any specific problem.

SUMMARY

[0006] In one aspect, a damping ring assembly for a drive sprocket of a mobile industrial machine, the damping member including a plurality of damping members, with each of the damping members including an arc length, a radial thickness, a plurality of recesses located on an end surface and a retention groove on an outer surface of the damping members. The damping ring assembly also includes a plurality of brackets, the plurality of brackets each including a radially inner leg, a radially outer leg, and a joining central portion that connects the radially inner leg and the radially outer leg.

[0007] In another aspect, a drive sprocket damping assembly for a mobile industrial machine, the drive sprocket damping assembly including a plurality of damping members, each of the damping members including an arc length, a radial thickness, and a plurality of recesses spaced along an end surface. Each of the plurality of recesses may include a threaded insert configured to receive a threaded fastener for securing individual damping member to a sprocket drum. Further, the damping assembly includes a plurality of brackets configured to be coupled to an end face of the sprocket drum at a first bracket end and to one of the threaded inserts in the end surface of the individual damping member at a second bracket end. The plurality of damping members together form a circumferential shape when installed onto a surface of a drive sprocket.

[0008] In yet another aspect, a method of assembling a damping ring assembly to a drive sprocket, the damping ring assembly including a plurality of damping members with a plurality of recesses located on an end surface and a plurality of brackets. The method includes fastening a radially inner leg of each of the plurality of brackets to the drive sprocket and fastening a radially outer leg of each of the plurality of brackets to one of the plurality of recesses located in the end surface of

the damping members. The step of fastening the radially inner legs and radially outer legs is repeated until an entire outer circumference of a surface portion of the drive sprocket is covered with a damping member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

[0010] FIG. 1 is a side view of a track-type mobile industrial machine including a sprocket damping assembly, according to aspects of the disclosure.

[0011] FIG. 2 is an enlarged, isometric view of the sprocket damping assembly on the drive sprocket of FIG. 1.

[0012] FIG. 3 is a cross-sectional view taken through line 3-3 of FIG. 2 depicting a portion of the drive sprocket and sprocket damping assembly.

[0013] FIG. 4 is a cross-sectional view taken through line 4-4 of FIG. 3 depicting a portion of the drive sprocket and sprocket damping assembly.

[0014] FIG. 5A an isometric, inner view of a section of a damping ring of the sprocket damping assembly of FIG. 3

[0015] FIG. 5B is an isometric outer view of a section of the damping ring of the sprocket damping assembly of FIG. 3.

[0016] FIG. 5C is an isometric view of a damping ring bracket of the sprocket damping assembly of FIG. 3.

DETAILED DESCRIPTION

[0017] Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. In this disclosure, unless stated otherwise, relative terms, such as, for example, “about,” “substantially,” and “approximately” are used to indicate a possible variation of $\pm 10\%$ in the stated value.

[0018] FIGS. 1 and 2 depict a mobile industrial machine 10 comprising a track drive assembly 12 with a power source (not shown), a continuous ground-engaging track 20, at least one drive sprocket 50, and a sprocket damping assembly 100. The sprocket damping assembly 100 (best shown in FIG. 2) includes a plurality of damping ring members 110 attached to the drive sprocket 50 via a plurality of retaining brackets 170 and axial fasteners, according to aspects of the present disclosure. While the mobile industrial machine 10 of FIG. 1 is shown in the context of a heavy-duty track-type tractor, the sprocket damping assembly 100 of the present disclosure is not thereby limited, and other types of track-type work machines, such as excavators, track loaders, and other similar machines, are within the scope of the present disclosure. As shown in FIG. 1, track-type tractor 10 further includes a tractor frame 14 mounted on top of the track drive assembly 12, an implement system 16 attached to a portion of the tractor frame 14, and an operator cabin 18.

[0019] The track drive assembly 12 includes the pair of endless or continuous ground-engaging tracks 20, a plurality of idler wheels 40-44, and the at least one drive sprocket 50. The ground-engaging tracks 20 (one of which is visible in FIG. 1) are mounted onto the drive sprocket 50 and idler wheels 40-44 of the track drive assembly 12. Note that while only a single drive sprocket 50, a single front idler 40, a single rear idler 42, and multiple rollers 44 have been shown in FIG. 1, track

drive assembly **12** may include different configurations. For example, FIG. **1** illustrates a high-drive configuration, where the drive sprocket **50** is positioned vertically higher than a front idler wheel **44** and a rear idler wheel **46**; however, alternative configurations, such as a low-drive configuration, are possible. The sprocket damping assemblies **100**, as discussed, may be implemented on the drive sprocket **50**, however, the same or similar damping assemblies **100** could be included additionally or alternatively on the front or rear idler wheels **40**, **42**, and/or the idle rollers **44** of the track-type tractor **10**.

[0020] FIG. **2** illustrates the drive sprocket **50** with the mounted sprocket damping assembly **100** and a portion of the ground-engaging track **20** in further detail. The drive sprocket **50** is a sprocket wheel that includes a sprocket drum **56** with an external circumferential surface **58** (located beneath a plurality of damping ring members **110** and labeled in FIG. **3**), a plurality of sprocket gear ring segments **52** mounted onto the sprocket drum **56**, and an end cap **72** secured to an axially outer end face of the sprocket drum **56**. Alternatively, the drive sprocket **50** may include a unitary sprocket gear ring rather than ring segments **52**. FIG. **2** also depicts a portion of the ground-engaging track **20** (track shoes omitted), and includes a plurality of track links **22** connected via link pins **24**. Each of the track links **22** includes a first link rail **26** and a second link rail **28** spaced apart by a link bushing **30**. Additional or alternative track components may be used in the ground-engaging track **20**. The spacing between the link rails **26**, **28** allows sprocket teeth **54** of the sprocket gear ring segments **52** to rotate between the link rails **26**, **28** as the track links **22** travel relative to the drive sprocket **50**. The track links **22** and the circumferential spacing of the sprocket teeth **54** are sized so that adjacent link bushings **30** may be received into grooves defined between adjacent sprocket teeth **54** to drive the ground-engaging track **20**. While the link rails **26**, **28** are spaced from the circumferential surface **58** of the sprocket drum **56** as the track links **22** travel about the drive sprocket **50**, the link bushings **30** contact the sprocket teeth **54** and are received within grooves between the adjacent sprocket teeth **54**.

[0021] The sprocket damping assembly **100** is installed onto the circumferential surface **58** of the sprocket drum **56**, as shown in FIGS. **2-4**. The sprocket damping assembly **100** includes the plurality of damping ring members **110**, an annular retention groove **140** located on a radially external surface **122** of each damping ring member **110**, and a member retention system that includes a retention device **160** and a plurality of retaining brackets **170**. As shown in FIG. **3**, each damping ring member **110** includes a main body **112** with a radially internal surface **114** and the radially external surface **122**, with the radial surfaces **114** and **122** defining a general radial thickness of the main body **112** (also shown in FIG. **4**). The damping ring members **110** may be fabricated from an elastomeric or resilient material, such as rubber, polyurethane, other polymers and elastomers and the like. The damping ring members **110** comprise an arcuate shape (shown in FIGS. **5A** & **5B**) and are assembled along the circumferential surface **58** of the sprocket drum **56**. For example, the sprocket damping assembly **100** of FIG. **2** includes four arcuate damping ring members **110** (three of which are visible in FIG. **2**); however, the sprocket damping assembly **100** may include additional or less damping ring members **110**. For example, in an alternative embodiment, the sprocket damping assembly **100** may be made of a single unitary damping ring.

[0022] When assembled or installed, the damping ring members **110** together comprise a generally annular shape that is complimentary to the shape of the circumferential surface **58** of the sprocket drum **56**. For example, in FIGS. **3** and **4**, the circumferential surface **58** of the sprocket drum **56** is shown to include a first constant-diameter surface **60** located proximal to the sprocket gear ring segments **52**, a tapered surface **62**, and a second constant-diameter surface **64** that is distal to the sprocket gear ring segments **52** and proximal to the sprocket drum end cap **72**. The tapered surface **62** slopes at a tapered angle ranging from 0 to 90 degrees, as measured relative to the first constant-diameter surface **60** or the second constant-diameter surface **64** from an exterior perspective. The first and second constant-diameter surfaces **60**, **64** may be substantially parallel to each other. It should be noted that, while FIGS. **3** and **4** illustrate the use of a circumferential surface **58** with a

pair of constant-diameter sections **60**, **64** and a tapered section **62**, the circumferential surface **58** may alternatively include a substantially uniform or constant diameter across the length of the sprocket drum **56**, or a diameter that otherwise varies along the length of the sprocket drum **56**. [0023] The radially internal surface **114** of the damping ring member **110**, as shown in FIGS. **3** and **4**, may have a conforming or matching shape with the various portions of the circumferential surface **58** of the sprocket drum **56**. Thus, the radially internal surface **114** of the damping ring member **110** may include a first radially inner constant-diameter section **116** that corresponds to the constant-diameter surface **60** of the sprocket drum **56**, a tapered section **118** of the damping ring member **110** that corresponds to the tapered surface **62**, and a second radially inner constant-diameter section **120** that corresponds to the second constant-diameter surface **64**. With this corresponding or matching shape between the circumferential surface **58** of the sprocket drum **56** and the radially internal surface **114** of an individual damping ring member **110**, the radially internal surface **114** of the damping ring member **110** conforms to and generally aligns with the circumferential surface **58** of the sprocket drum **56**. Further, the tapered surface **62** and tapered section **118** taper radially inwardly away from a second axially outer end surface **128** (shown in FIG. **4**), thereby restricting the damping ring members **110** from moving axially away from the sprocket gear ring segments **52**. As used in this disclosure, the phrase constant-diameter or tapered means a substantially or approximately constant diameter or tapered, respectively.

[0024] Referring to FIG. **4**, the radially external surface **122** of the damping ring member **110** may include a third radially outer constant-diameter section **126** and the retention groove **140**. The third radially outer constant-diameter section **126**, which generally corresponds to and is parallel with the first radially inner constant-diameter section **116** of the radially internal surface **114** of the damping ring member **110**, includes a generally constant radial thickness $t_{sub.c}$ of the damping ring member **110** which may be based on a plurality of factors, including the assembled height of the drive sprocket **50**, the relative lifespan of the drive sprocket **50**, optimized compression calculations for reducing potential gaps between the drive sprocket **50** and the link rails **26**, **28**, and other similar data points. The third radially outer constant-diameter section **126** is bounded between a first axially inner end surface **124** of the damping ring member **110** and a corner **141** that radially extends into a first retention wall **142** of the retention groove **140**. The corner **141** may be rounded or filleted (shown in FIG. **4**) or may be a 90 degree angle (not shown) relative to the radially external surface **122** and the first retention wall **142**.

[0025] The retention groove **140**, as shown in FIG. **4**, is bounded on either side by the first retention wall **142** and a second retention wall **146**, which may include the same height as, a height that is less than, or a height that is greater than the third radially outer constant-diameter section **126**. For example, in FIG. **4**, the heights of the of the first and second retention walls **142**, **146** are shown to be radially elevated above the third radially outer constant-diameter section **126** of the external surface **122**. The second retention wall **146** opposes the first retention wall **142** and terminates in the second axially outer end face **128** of the damping ring member **110**. The annular retention groove **140** includes an axial width $w_{sub.RG}$ that is less than a general width of the individual damping ring members **110** (shown in FIG. **4**). Each of the first and the second retention walls **142**, **146** include an axial width, $w_{sub.1R}$ and $w_{sub.2R}$ respectively. In FIG. **4**, the axial width $w_{sub.2R}$ of the second retention wall **146** is shown to be greater than the axial width $w_{sub.1R}$ of the first retention wall **142**, such that the axial width of the second retention wall $w_{sub.2R}$ is capable of retaining a plurality of bracket threaded inserts **136** within the second retention wall **146**; however, the axial widths, $w_{sub.1R}$ and $w_{sub.2R}$, may vary based on the specific geometry and overall dimensions of the drive sprocket **50**. As depicted in FIG. **4**, the retention groove **140** is recessed radially inward from an exterior perspective, toward the radially internal surface **114** of the damping ring member **110**, and has a variable radial thickness $t_{sub.RG}$ (based on tapering section **118**) that is less than the constant radial thickness of the damping ring member $t_{sub.c}$. In an alternative embodiment, the retention groove radial thickness $t_{sub.RG}$ may

be substantially equal to or greater than the constant radial thickness $t_{\text{sub.c}}$ of the third radially outer constant-diameter section **126**. The retention groove radial thickness $t_{\text{sub.RG}}$ is sized such that a retention device **160** is wholly retained within the retention groove **140**.

[0026] In FIGS. **4** and **5A**, each of the damping ring member **110** include an arc length $l_{\text{sub.DM}}$ (FIG. **5A**) and one or more indexing cavities **130** circumferentially spaced along the first axially inner end surface **124** of the damping ring member **110**. Each of the one or more indexing cavities **130** include a cutout portion **132** that extends partially through the main body **112** from the radially external surface **122** and terminates within the main body **112** of the damping ring member **110**. The cutout portions **132** may include an axial width $w_{\text{sub.IC}}$ (as measured from the first axially inner end surface **124** and extending axially outward towards the second axially outer end surface **128**) and a radial thickness $t_{\text{sub.IC}}$ (as measured from the first radially inner constant-diameter section **116** and extending radially outward towards the third radially outer constant-diameter section **126**). As shown in FIG. **4**, the cutout portions **132** may be sized such that the indexing cavities **130** are capable of receiving both an axial width and a radial thickness of a head of a bolt **55** that is used to secure the sprocket gear ring segment **52** to the sprocket drum **56**. Additionally, the cutout portions **132** are arranged along the arc length $l_{\text{sub.DM}}$ of the first axially inner end surface **124** in a spaced configuration, such that one indexing cavity **130** may receive a single bolt head **55**, thereby assisting in restricting rotation of the damping ring members **110**. For example, in FIG. **5A**, the axial width $w_{\text{sub.IC}}$ and the radial thickness $t_{\text{sub.IC}}$ of the indexing cavities **130** may be sized to retain a single bolt head **55**; however, indexing cavities may be sized to retain more than one bolt head **55**.

[0027] As noted above, the sprocket damping assembly **100** includes a member retention system that comprises the retention device **160** and the plurality of retaining brackets **170** (shown in FIGS. **3**, **4**, and **5C**). The retention device **160** provides compression to assist in restricting axial, radial, and rotational movement of the damping ring members **110** about the sprocket drum **56** when the retention device **160** is installed into the retention groove **140** (shown in FIGS. **3** and **4**). The retention device **160** may take the form of a band clamp, utilizing either a single or multiple clamp sections, with a tension adjusting mechanism (not shown). The retention device **160** may be made from one or more thin strips of steel, other metals or other appropriate materials that may be flexible enough to conform to the external surface **122** of the damping ring members **110** within the retention groove **140**, while simultaneously being resistant to stretching.

[0028] Additionally, as shown in FIGS. **3**, **4**, and **5C**, the plurality of damping ring members **110** utilize retaining brackets **170**, such as a two-hole “Z” type bracket, to connect to the sprocket drum **56** (via an individual threaded drum recess **68**) at a first end **171** of the retaining bracket **170** and to the damping ring member **110** at a second end **179** of the bracket. The retaining brackets **170** include a radially inner leg **172**, a radially outer leg **178**, and a joining central portion **176**. Shown in FIGS. **4** and **5C**, the radially inner leg **172** and the radially outer leg **178** are substantially parallel, while the joining central portion **176** of the retaining bracket **170** is substantially perpendicular to both the radially inner leg **172** and the radially outer leg **178**. The second axially outer end face **128** of an individual damping ring member **110** is offset, both axially and radially, from the sprocket drum end cap **72** in an axially inward and radially outward direction from the sprocket drum cap **72**. As such, when the damping assembly **100** is assembled on the drive sprocket **50**, the first end **171** of the individual bracket **170** is axially and radially distanced from the second end **179** of the bracket **170**, with the joining central portion **176** connecting or spanning the distance between the two structures.

[0029] Referring back to FIG. **4**, the sprocket drum **56** includes a plurality of threaded drum recesses or blind holes **68** (one of which is shown in FIG. **4**) that are circumferentially spaced along an axially outer end face **66** of the drum **56**. The end cap **72**, which includes a plurality of endcap through holes **74** (one of which is shown in FIG. **4**), is mechanically fastened to the sprocket drum **56** along the axially outer end face **66** by inserting a plurality of threaded drum bolts **70** through an

endcap through hole **74** and into an individual threaded drum recess **68**. As best shown in FIGS. **4** and **5C**, the first end **171** of the radially inner leg **172** includes a circular through hole **174**, through which the individual threaded bolt **70** is inserted to attach the radially inner leg **172** to the endcap **72**.

[0030] In FIG. **5B**, the second axially outer end face **128** of the damping ring members **110** includes a plurality of connection recesses **134** that extend in an axially inward direction from the end face **128**. Each connection recess **134** receives a corresponding threaded insert **136**, which may be attached during the molding process (e.g., overmolding) to maintain the integrity of the damping ring member **110** material or may be mechanically attached via insertion. Each threaded insert **136** includes an external profile **137** that securely fastens the individual insert **136** within an individual connection recess **134** of the damping ring member **110** and an internal threaded profile **139** that corresponds to a bracket fastener **138** (e.g. a threaded bolt) for attaching the second end **179** of the radially outer leg **178**, through a substantially ovular or slot-shaped through hole **180**, to the damping ring members **110**. The retaining bracket **170** is securely attached to the sprocket drum **56** (via the threaded drum fastener **70**) and to the damping ring member **110** (via the threaded bracket fastener **138**), as depicted in FIG. **4**. Alternatively, the retaining bracket **170** may include a plurality of washers (not shown), to be used in combination with the threaded bracket fasteners **138** and the threaded inserts **136**, to fasten the retaining bracket **170** to the damping ring member **110**.

INDUSTRIAL APPLICABILITY

[0031] The disclosed aspects of the sprocket damping assembly **100** for the drive sprocket **50** of the present disclosure may be used reduce the amount of noise generated by the metal-to-metal contact of the sprocket teeth **54** coming into contact with the bushings **30** of the track links **22** when the tractor-type industrial machine **10** is operating. Additionally, the sprocket damping assembly **100** may assist in limiting axial and radial movement of a mounted damping ring members **110** during operation of the machine **10**.

[0032] As shown in FIGS. **1** and **2**, the sprocket damping assembly **100** is configured to contact the track links **22** during engagement of the bushings **30** with the sprocket teeth **54**. In order to do so, the sprocket damping assembly **100** is installed onto a circumferential surface **58** of the sprocket drum **56**. The sprocket damping assembly **100** includes a plurality of damping ring members **110** that are installed onto the circumferential surface **58**, such that the circumferential surface **58** of the sprocket drum **56** is completely enveloped. Each damping ring members **110** is installed by aligning a plurality of indexing cavities **130** with a plurality of sprocket gear member bolts **55** that attach segments of a sprocket gear ring **54** to the sprocket drum **56**, as shown in FIGS. **3** and **4**. The plurality of bolts **55** from the sprocket gear ring segment **52** are placed into cutout portions **132** in a first axially inner end face **124** of the damping ring members **110**, thereby securing a first portion of the individual damping ring members **110** and assisting in preventing rotational movement of the damping ring members **110**. The individual damping ring members **110** include a plurality of radially inner surfaces **116-120** that conform to the external surfaces **60-64** of the circumferential surface **58** of the sprocket drum **56**, as illustrated in FIGS. **3** and **4**.

[0033] Once the first axially inner end face **124** of the damping ring member **110** has been secured, a plurality of retaining brackets **170** are attached or secured to an end cap **72** of the sprocket drum **56** (via a plurality of threaded drum recesses **68**) in an axially outer end face **66** of the sprocket drum **56**. The plurality of retaining brackets **170** each include a radially inner leg **172** with a circular through hole **174** (shown in FIGS. **3**, **4**, and **5C**). Individual threaded drum fasteners **70** are inserted through the circular through hole **174** of the retaining bracket **170**, through an end cap through hole **74**, and into an individual threaded drum recess **68**, thereby securing the radially inner leg **172** of the retaining bracket **170** to the drive sprocket **50**. A radially outer leg **178** of the retaining bracket **170**, which includes a substantially ovular or slot-shaped through hole **180**, is used to connect the retaining bracket **170** to the damping ring member **110**. Specifically, an individual threaded bracket fastener **138** (e.g. a threaded bolt) is inserted through the ovular or slot-

shaped through hole **180** and into an internal threaded profile **139** of a threaded bracket insert **136** that has installed into a bracket connection recess **134** on the second axially outer end face **128**. The oval through hole **180** is configured to allow for the adjustment of the retaining bracket **170** during attachment to the damping ring member **110** when the retaining bracket **170** is securely attached to the sprocket drum **56**. As shown in FIG. 5B, an exemplary damping ring member **110** is shown to include four threaded bracket inserts **136** per member **110**; however, the number of threaded bracket inserts **136** installed onto the damping ring member **110** may vary depending upon the needs of the user. The installation of the damping ring members **110** is repeated until the damping ring members **110** form an annular shape and the circumferential surface **58** of the sprocket drum **56** is fully encapsulated, as shown in exemplary FIG. 2. It should be noted, however, that the user may instead choose to strategically space the damping ring members **110** apart from one another in order to develop zones of protection, as also contemplated by the inventor.

[0034] Once the plurality of damping ring members **110** has been installed or circumferentially coupled onto the circumferential surface **58** of the sprocket drum **56**, a retention device **160** is installed into the retention groove **140**, as shown in FIG. 4. The retention device **160** may provide additional support to the sprocket damping assembly **100** by securing a portion of the damping ring members **110** at another point along their length.

[0035] While the installation of the sprocket damping assembly **100** is shown to be on an axially outer side of the drive sprocket **50** as shown in exemplary FIGS. 2 and 3, the sprocket damping assembly **100** may be installed on either side of the sprocket gear ring **54** depending on the configuration of a particular drive sprocket **50** on which the sprocket damping assembly **100** is implemented. Furthermore, those skilled in the art would understand that similar sprocket damping assemblies **100** may be installed on other sprocket wheels supporting the ground-engaging track **20**, such as the idler gears, rollers and the like.

[0036] In accordance with the present disclosure, the sprocket damping assembly **100** may assist in reducing noise generated by the metal-to-metal contact of the sprocket teeth **54** and the ground-engaging track **20** during operation. Additionally, by securing the damping ring members **110** at both a first axially inner end face **124** (via the drive sprocket bolts **55**) and a second axially outer end face **128** (via the retention or “Z” brackets **170**), the sprocket damping assembly **100** of the present disclosure helps to ensure that the assembly **100** will contact the track **20** and reduce any potential axial, radial, or rotational movement of the damping rings **110**. The sprocket damping assembly **100** is therefore able to incorporate a simple design that utilizes the existing fasteners (e.g., drive sprocket bolts **55** and threaded drum fasteners **70**) of the drive sprocket **50** for additional efficiency. Finally, the inclusion of the damping ring members **110** for contacting the track **20** prior to the sprocket teeth **54** may assist in the reduction of forces imparted onto the sprocket teeth **54**, thereby decreasing the amount of wear and strain on the sprocket teeth and potentially resulting in improved maintenance costs.

[0037] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed system without departing from the scope of the disclosure. Other embodiments of the system will be apparent to those skilled in the art from consideration of the specification and practice of the system disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A damping ring assembly for a drive sprocket of a mobile industrial machine, the damping member comprising: a plurality of damping members, each of the damping members including an arc length, a radial thickness, a plurality of recesses located on an end surface and a retention groove on an outer surface of the damping members; and a plurality of brackets, the plurality of

brackets each including a radially inner leg, a radially outer leg, and a joining central portion that connects the radially inner leg and the radially outer leg.

2. The damping ring assembly of claim 1, wherein the plurality of brackets are configured to be coupled to an end face of the drive sprocket at the radially inner leg and to one of the plurality of recesses in the end surface of an individual damping member at the radially outer leg.

3. The damping ring assembly of claim 2, wherein the radially inner leg and the radially outer leg extend substantially parallel to each other.

4. The damping ring assembly of claim 3, wherein the joining central portion is substantially perpendicular to both the radially inner leg and the radially outer leg.

5. The damping ring assembly of claim 4, wherein the radially inner leg includes a circular through hole configured to connect the radially inner leg to the end face of the sprocket drum via a mechanical fastener.

6. The damping ring assembly of claim 5, wherein the radially outer leg includes a through hole configured to connect the radially outer leg to a recess of the plurality of recesses of the individual damping member via a threaded mechanical fastener.

7. The damping ring assembly of claim 1, wherein the retention groove includes an annular groove width that is less than a width of an individual damping member.

8. The damping ring assembly of claim 1, wherein each damping member includes an internal surface having a first substantially constant-diameter section, a substantially tapered section, and a second substantially constant-diameter section.

9. The damping ring assembly of claim 1, wherein each of the plurality of recesses includes a threaded insert configured to receive a threaded fastener and to secure the individual damping member to a sprocket drum.

10. A drive sprocket damping assembly for a mobile industrial machine, the drive sprocket damping assembly comprising: a plurality of damping members, each of the damping members including an arc length, a radial thickness, and a plurality of recesses spaced along an end surface, wherein each of the plurality of recesses includes a threaded insert configured to receive a threaded fastener for securing individual damping member to a sprocket drum; and a plurality of brackets configured to be coupled to an end face of the sprocket drum at a first bracket end and to one of the threaded inserts in the end surface of the individual damping member at a second bracket end, wherein the plurality of damping members together form a circumferential shape when installed onto a surface of a drive sprocket.

11. The damping ring assembly of claim 10, wherein each of the plurality of brackets include a radially inner leg, a radially outer leg, and a joining central portion that connects the radially inner leg and the radially outer leg.

12. The damping ring assembly of claim 11, wherein the radially inner leg and the radially outer leg are substantially parallel to each other, and wherein the radially inner leg and the joining central portion are substantially perpendicular.

13. The damping ring assembly of claim 11, wherein the radially inner leg includes a circular through hole configured to connect the radially inner leg to the end face of the sprocket drum via a mechanical fastener.

14. The damping ring assembly of claim 11, wherein the radially inner leg includes an ovalar through hole configured to connect the radially outer leg to the plurality of recesses of the individual damping member via a threaded mechanical fastener.

15. The damping ring assembly of claim 14, wherein the ovalar through hole is configured to allow for the adjustment of the individual bracket relative to an individual hole of the individual damping member.

16. The damping ring assembly of claim 10, further including a plurality of cutout portions circumferentially spaced about an opposite end surface of the damping member with respect to the end surface having the plurality of recesses.

17. The damping ring assembly of claim 10, wherein each damping member includes an internal surface including a first substantially constant-diameter section, a substantially tapered section, and a second substantially constant-diameter section.

18. The damping ring assembly of claim 17, wherein the substantially tapered section tapers radially inwardly away from the end surface having the plurality of recesses.

19. The damping ring assembly of claim 18, wherein, in an assembled configured, the first bracket end is located axially and radially distanced from the second bracket end.

20. A method of assembling a damping ring assembly to a drive sprocket, the damping ring assembly including a plurality of damping members, each of the damping members including an arc length, a radial thickness, a plurality of recesses located on an end surface, and a retention groove on an outer surface of the damping members; and a plurality of brackets, the plurality of brackets each including a radially inner leg, a radially outer leg, and a joining central portion that connects the radially inner leg and the radially outer leg, the method comprising: fastening the radially inner leg of each of the plurality of brackets to the drive sprocket; and fastening the radially outer leg of each of the plurality of brackets to one of the plurality of recesses located in the end surface of the damping members, wherein the fastening of the radially inner legs and radially outer legs is repeated until an entire outer circumference of a surface portion of the drive sprocket is covered with a damping member.
